

7(a)	300 Ω	A1
(b)(i)	$\frac{R}{300} = \frac{6.0-2.4}{2.4} \text{ or any other method}$ $R = 450 \Omega$	C1 A1
(b)(ii)	Resistance of the LDR increases Since current drop, pd across fixed resistor drops, OR using potential divider, potential difference across the LDR increases [B1]	B1 B1
(c)	$R = V/I = 3.0 / 40 \times 10^{-3} = 75 \Omega$	A1 A1
(d)(1)(2)	Fig. 7.4	B1 A1

	L: straight line passing through (0,0) and (6,20) Note: The LDR follows Ohm's Law and has a resistance of $300\ \Omega$ X: same as L between $pd = 0\text{ V}$ and 2 V, parallel to LED graph above 2 V Component X will not allow current in the LED branch but will allow the same current in the LDR branch below 2 V. Above 2 V, current in component X = sum of current in LDR and LED (accurately, we should have a curve that starts parallel to LED graph but eventually curve away from it, but this effect is not significant over this range.)	
d)(1)(2)	$9.0 \pm 0.5\text{ mA}$ $9 \pm 1\text{ mA}$ Find the current where the pd across X and fixed resistor ($450\ \Omega$) sums to 6 V. (Note: In a series circuit, the two components in series will have the same current flowing through them.) You will need to draw the line for $450\ \Omega$ resistance first though. Pro-tip: Place a ruler parallel to x-axis and shift it up/down.	A2 A1

